

Miguel Beltran

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EDUCATION

University of California, Davis

Bachelor of Science in Computer Science

Davis, CA

Expected June 2027

Relevant Coursework: Algorithms & Data Structures, Operating Systems, Machine Learning, Artificial Intelligence, Computer Architecture

EXPERIENCE

ASUCD Picnic Day

Assistant Technical Director

Davis, CA

Nov 2025 - Present

- Contributed to a React/TypeScript event platform supporting 70,000+ attendees during UC Davis's annual Picnic Day event.
- Built in-app onboarding tutorials, schedule exports, and interactive map features to improve attendee navigation and accessibility.
- Migrated application data infrastructure from Supabase to embedded SQLite with authenticated Google Sheets synchronization, implementing upserts and stale-record cleanup.
- Containerized the full-stack application with Docker Compose, reducing onboarding to one `docker compose up` command and improving environment consistency.

Include

Software Engineer

Davis, CA

Oct 2025 - Present

- Developed 3 responsive React/Next.js components for the Space Systems Services client website, including "Who We Are", "Why Join SSS", and "Mission" sections.
- Built modular SCSS styling with reusable responsive mixins, optimized Next.js image rendering, and prop-driven content layouts for maintainable client-facing pages.

Google Developer Student Club

Software Engineer

Davis, CA

Oct 2024 - Jun 2025

- Built a Slack bot using Python, Flask, ngrok, and the Slack API to automate community workflows and onboarding support.
- Developed an announcement automation system using Slack slash commands and parameter parsing, enabling organizers to schedule formatted announcements directly from chat commands and saving 2+ hours/week.

PROJECTS

dltracker - Deadlock Analytics Platform | Next.js 16, TypeScript, PostgreSQL, Redis, Playwright, Vitest, GitHub Actions

- Built a full-stack analytics platform used by 1000+ users, integrating Steam and Deadlock APIs across 10 pages and 45+ reusable components.
- Designed a 3-signal identity disambiguation system resolving up to 100 ambiguous player records per leaderboard page across 5 regions.
- Implemented GitHub Actions CI/CD, Playwright/Vitest testing, Redis-backed rate limiting, Zod validation, Sentry monitoring, and ISR caching for reliability and performance.

Song Popularity Predictor - ML Pipeline | Python, TensorFlow, XGBoost, scikit-learn

- Designed and evaluated a supervised ML pipeline across 32,000+ Spotify tracks using Random Forest, XGBoost, and Logistic Regression models.
- Reformulated low-signal regression tasks into binary hit classification after feature analysis.
- Engineered artist popularity and genre features with GridSearchCV tuning, improving Random Forest accuracy to 80%.

Kitch - Kitchen Management App | Java, Spring Boot, React, H2, JPA

- Built a full-stack kitchen management app with a Spring Boot REST API and React/Vite frontend for inventory, shopping lists, and recipes.
- Designed layered backend architecture with controllers, services, repositories, DTOs, and JPA entities across 20+ REST endpoints.

8-Bit CPU - Custom Processor | Logisim, Computer Architecture

- Designed a 5-stage pipelined 8-bit CPU with hazard detection and data forwarding, executing 20+ custom ISA instructions.

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, Java, C, C++, Go, SQL

Frontend: React, Next.js, Tailwind CSS, SCSS, HTML/CSS, Framer Motion

Backend: Node.js, Express, Spring Boot, Flask, FastAPI, REST APIs, Prisma

Databases & Caching: PostgreSQL, SQLite, Redis, Firestore, Supabase

ML & Data: TensorFlow, PyTorch, XGBoost, scikit-learn, Pandas, NumPy

Testing: Playwright, Vitest, Pytest

DevOps & Infrastructure: Docker, GitHub Actions, CI/CD, Linux, Vercel, Sentry